Title: Tropical Cyclone Wind Intensity Drives Phase-Specific Displacement of Emergency Department Demand in Florida, U.S., 2004–2022

Recommendation: Major revision

Summary:

This manuscript examines phase-specific changes in emergency department (ED) utilization associated with TCs in Florida during 2004–2022. The topic is interesting and potentially relevant to healthcare preparedness before and after TC impacts. However, several issues regarding storm exposure classification, the treatment of suppressed ED counts, statistical uncertainty, and the interpretation of intensity-dependent effects need to be addressed. The distinction between changes in ED utilization and displacement of healthcare demand also requires clarification. I therefore recommend Major Revision.

General Comments:

1. Lines 94–98: Please clarify whether the Impact day is defined by Florida landfall, closest approach to each ZIP code, or another criterion. These definitions may assign different dates to the same location. The choice of eight-day Threat and Cleanup windows also requires justification. Daily estimates around the impact date and sensitivity analyses using alternative window lengths would help show whether the phase averages conceal substantial temporal variation. The treatment of overlapping storm windows should also be assessed.
2. Lines 93–103: The limited number of TC events raises concerns about the robustness of the results, particularly after intensity stratification, with only 14 storms in the ≥64 kt subset. Why is the analysis restricted to 2004–2022? Please explain why data before 2004 were not included and whether the analysis could be extended through 2025 to increase the sample size. If these temporal limits reflect ED data availability, changes in reporting practices, or data comparability, please clarify these constraints. A longer record, where comparable data are available, would help assess whether the phase-specific patterns and their intensity dependence are robust or disproportionately influenced by a few major storms.
3. Lines 92–103 and 245–249: The definition of TC exposure requires further clarification. Storm swaths are constructed using the radius of maximum wind, whereas the analyses are described in terms of wind thresholds of ≥34, ≥55, and ≥64 kt. A swath based on the radius of maximum wind does not directly delineate the area experiencing each wind threshold. Please specify the IBTrACS variables used, the treatment of missing radius data, the temporal interpolation procedure, and how ZIP codes were classified as exposed. A sensitivity analysis using available wind radii or modeled local wind exposure would help assess the robustness of the results. A map of storm tracks, exposure swaths, and included ZIP codes would also be useful.
4. Lines 100–103 and 118–121: Please provide a table listing the 44 storms, their Florida impact dates, the wind speeds used for classification, and the numbers of exposed ZIP codes. It is unclear whether intensity refers to lifetime maximum intensity, intensity near Florida, or intensity at local exposure. This distinction is particularly important for storms that reached major hurricane intensity away from Florida. The rationale for selecting 55 kt should also be explained.
5. Lines 108–117: The treatment of suppressed ED counts is insufficiently described. The logistic component estimates the probability of a true count ≤6, and the Poisson component models counts >6. However, the manuscript does not explain how the expected count within the suppressed group is estimated when combining these components into unconditional expected

ED visits. Please provide the model equations and prediction formula, specify whether the Poisson component accounts for truncation at six, and report the proportion of suppressed observations by storm phase. Sensitivity to alternative treatments of suppressed counts should be assessed.

6. Lines 159–170: The three wind thresholds define nested samples rather than independent intensity categories. Stronger storms contribute to multiple analyses, and the exposed ZIP codes and baseline utilization may differ across thresholds. Therefore, larger absolute changes in visits do not by themselves establish an intensity-dependent response. Please report the baseline expected visits and relative percentage changes for each threshold, and assess intensity differences using a common target population or a model with phase-by-intensity interactions.
7. Lines 174–178: The confidence intervals are obtained from ZIP-clustered bootstrap resamples. However, multiple ZIP codes are affected by the same storm, and their responses may be correlated. Resampling ZIP codes alone may not fully capture uncertainty associated with the limited number of storm events, particularly the 14 hurricanes. Please justify the resampling design and assess robustness using an approach that accounts for dependence within storms. A leave-one-storm-out analysis would also help determine whether a few major events dominate the findings.
8. Lines 199–212 and 265–277: The proposed displacement of medical care from the Impact phase to the Cleanup phase is plausible, but the present analysis does not directly establish this mechanism. Post-storm increases could include new injuries or illnesses, while impact-day reductions could reflect evacuation, facility closures, or changes in reporting. Please clarify whether ZIP codes refer to patient residence or facility location and whether visits outside Florida are captured. The causal wording in the title, particularly "Drives," should be reconsidered, and deferred care should be presented as a possible explanation.
9. Lines 133–148 and 180–197: The placebo analyses are mainly reported as coefficients from the separate logistic and Poisson components, whereas the main findings are expressed as combined differences in expected visits. Please report the placebo results on the same marginal-effect scale as the main analysis and show the observed estimates relative to the randomization distributions. In addition, a minimum separation of 16 days does not necessarily eliminate overlap between two inclusive ±8-day windows. Please clarify whether overlap with the true exposure windows, including those of other storms, was excluded.
10. Figure 1: Please improve the figure resolution and label readability. Panel headings should include the ≥ symbol to make clear that these are cumulative thresholds. The annotation within the figure describes the points as "bootstrap means," whereas the caption describes them as "point estimates." Please clarify which estimates are plotted and ensure consistency among the figure, caption, and reported numerical results.

Minor formatting and grammatical revisions are proposed as follows:

(1) Lines 45–47: "these analyses are primarily through" → "these analyses are primarily based on."

(2) Lines 69–71: The sentence beginning "Spatial resolution at the ZIP code level…" is incomplete because "between TC exposure" lacks a second object. Please clarify that the analysis aligns TC exposure with ED utilization.

(3) Lines 72–79: The Introduction could better distinguish anticipatory behavioral responses from evolving physical hazards before landfall. Recent research has shown that TC rain rates

increase as storms approach the coast, indicating that the pre-landfall period may also involve direct exposure to adverse weather. Briefly discussing this evidence would strengthen the rationale for the phase-specific analysis and provide a broader context for interpreting the Threat-phase increase in ED utilization. One relevant reference is:

Zhong, Q., Gan, J., Tu, S., Toumi, R., and Chan, J. C. L. (2026). Global increase in rain rate of tropical cyclones prior to landfall. Nature Communications, 17, 114. https://doi.org/10.1038/s41467-025-68070-z

(4) Lines 201 and 205: "on the Impact phase" → "during the Impact phase." In line 205, "decreases of ED utilization" should also be changed to "decreases in ED utilization."

(5) Lines 219–222: Please describe the changes in magnitude more precisely. The Impact-phase reduction increases by a factor of approximately 2.57, while the Cleanup-phase increase rises by a factor of approximately 2.97.

(6) Lines 225–227: Please revise the sentence beginning "Therefore, our results preserve…" to clarify the antecedent of "that" and correct the subject–verb agreement.

(7) Lines 291–293: "All analysis code … are available" → "All analysis code … is available."

(8) Please standardize the presentation of confidence intervals, particularly intervals with negative endpoints, and use consistent terminology for wind intensity and storm phases throughout the manuscript.